

M17

USER MANUAL



Please read this manual thoroughly before installing and using the instrument.

Statement

The information contained in this manual is based on the experience and knowledge gained by the Company in the product areas explored prior to the issuance of this manual.

Whilst the Company is satisfied that the information provided in this manual is accurate and reliable, the Company disclaims all warranties of any kind, including (but not limited to) the implied warranties of merchantability and fitness for a particular purpose. The Company shall not be liable for any loss or damage, including special damages arising out of or in connection with the use of this information, even if such loss or damage is caused by negligence or other error on the part of the Company.

The copyright of this manual is owned by the Company and its contents and information may not be reproduced or transmitted without the written permission of the Company.

The contents contained in the manual are subject to change without notice due to product upgrades and improvements.

Users are requested to read this manual in detail before installing and using the instrument.

Contents

CHAPTER 1: OVERVIEW	01
1.1 Product Features	01
1.2 Main Uses and Scope of Application	01
CHAPTER 2: PRECAUTIONS FOR SAFE USE	05
2.1 Safe use period	05
2.2 Safe Use	05
2.3 Prone to incorrect use or operation	05
2.4 Emergency Measures for Unusual Situations	05
2.5 Considerations for Special Situations	05
2.6 Contraindicated population	05
2.7 Symbols and labels	06
CHAPTER 3: STRUCTURE AND WORKING PRINCIPLE DIAGRAM	07
3.1 Working Principle	07
3.2 Workflow of main components or functional units	08
CHAPTER 4: TECHNICAL CHARACTERISTICS	08
CHAPTER 5: SIZE & WEIGHT	09
CHAPTER 6: INSTALLATION AND COMMISSIONING	09
6.1 Installation Method	09
6.2 Installation Precautions	09
6.3 Installation Test Items	09
6.4 Preparation Before Trial Operation & Trial Operation	09
CHAPTER 7: OPERATING GUIDE	09
7.1 Interface Overview	09
7.2 Data Upload and Download	13
7.3 File Management	14
7.4 Photo capture and analysis	16
7.5 Report Viewing	17
7.6 Analysis Compare	18
7.7 Magnify Analysis	20
7.8 3D Slice	20
7.9 Skin Aging	21
7.10 Product Management	21
7.11 Update	22
CHAPTER 8: TROUBLESHOOT	23
CHAPTER9. SAFETY DEVICES AND ACCIDENT HANDLING	23
9.1 Installation of protective devices and precautions	23
9.2 How to deal with accidents	23
9.3 Emergency handling methods	23
CHAPTER10. MAINTENANCE AND REPAIR	24
CHAPTER11. TRANSPORTATION AND STORAGE	24
CHAPTER12. UNPACKING AND INSPECTION	25
CHAPTER13. ENVIRONMENTAL AND OTHERS	25

Chapter 1: Overview

1.1 Product Features

1. This instrument adopts non-invasive technology, safe and non-invasive.
2. This instrument adopts Android system to take pictures, analyse and display all-in-one machine.
3. Using 2000W pixel high-definition camera
4. Big data AI calculation method to analyse RGB pores, RGB pigmentation, RGB texture, PL roughness, UV acne, UV pigmentation, UV roughness, red zone, brown zone. UV pigment, UV roughness, red zone, brown zone, black and white zone, skin aging, skin whitening.
5. Full face control analysis and local control analysis
6. Skin magnification, 3D imaging and skin slicing.
7. Cloud data storage
8. Hierarchical management of beauty organisations
9. Product brand management and product upload with one click

1.2 Main Uses and Scope of Application

1.2.1 Main purpose

Intelligent Facial Photographing Instrument is one of the facial skin image data statistics instruments, which can take high-definition images of the face under the RGB+UV+PL tri-spectrum and analyse the skin characteristics using artificial intelligence AI algorithm, thus providing data reference for the beauty industry personnel to customize skin care solutions, and at the same time with the skin magnifying glass, it can be more detailed observation of the local skin condition.

1.2.2 Scope of application

Mainly used in beauty salons and homes and other places to observe skin characteristics

1.2.3 Name and specification

Name: Intelligent Facial Photographing Instrument

Specifications: M17

1.2.4 Environment

1.2.4.1 Environmental requirements:

This instrument is suitable for indoor operation.

Ambient temperature range when the instrument is in operation: 10°C ~ 30°C.

Relative humidity range: 30% ~ 75%.

Atmospheric pressure range: 700hpa ~ 1060hpa

Ambient temperature range during storage: 0°C ~ 40

Relative humidity range: 30% ~ 90%

1.2.4.2 Power requirements:

Rated power supply: 110~230 VAC/50Hz~60Hz

Maximum power: 25W

1.2.5 Safety

Only trained and qualified service personnel authorised by the Company may open the housing of the equipment to replace electrical and mechanic

1.2.5.1 Mechanical safety

Warning:

Unless otherwise specified, the unit must not be operated with the chassis enclosure open.

1.2.5.2 Electrical safety

	Warning: Risk of electric shock - For the protection of the user and operator, ensure that the unit's enclosure is earthed and that the protective earth of the supply socket is intact. Do not connect the 3-wire cable of this device to a 2-wire plug!
	Warning: It should be ensured that the external earth wire of the equipment is connected correctly and securely, and the earth wire of the equipment should not be removed for any reason.
	Warning: Danger of electric shock - Do not open the unit's casing during operation or when the power is connected, only an authorised service engineer should do so.

	Be careful: Before use, the user must inspect the equipment and cables for any obvious damage that may affect the customer's safety or the instrument's performance. The recommended inspection cycle is once a week or less. If obvious damage is found, it is recommended that the damaged parts be replaced before use.
	Be careful: Switch off the system before cleaning. Cleaning consists of wiping dust from the surface of the unit with a soft brush or soft cloth, using a brush to sweep dust from connectors, panel edges, or using a non-corrosive cleaner such as dishwashing liquid. Do not allow the cleaner to penetrate the inside of the unit.
	Be careful: During the operation of the equipment, if there is an emergency, you can quickly cut off the power supply through the emergency stop switch on the side of the main body of the equipment to terminate the operation..

1.2.5.3 Environmental safety

	Warning: Do not operate this device outside of the operating environment limits, as this may cause the device to operate incorrectly. The operating environment limits are as follows: Operating temperature: 10°C to 30°C Relative humidity: 30% to 75% Atmospheric pressure range: 700hpa ~ 1060hpa Power supply voltage: 110~230 VAC Power frequency: 50Hz~60Hz Maximum power: 25W
	Be careful: Keep the environment clean and avoid shock. Keep away from corrosive drugs, dust, high temperature and humidity.

	Warning: Electromagnetic interference - Ensure that the instrument is installed and used in an environment free from strong electromagnetic interference, such as that caused by wireless transmitters or mobile phones.
	Warning: Do not use in the presence of explosive anaesthetics or other flammable substances.

1.2.5.4 Operational safety

	Warning: If you feel unwell when using this device, stop using it immediately.
	Warning: - If any of the following conditions are encountered, unplug the instrument from the power outlet and contact us promptly: - Liquid splashed on the instrument, drenching; - The instrument is dropped or the casing is broken; - The instrument emits odour, smoke or abnormal noise.
	Warning: Unplug the power supply after use to ensure safety.
	Warning: - Do not disassemble the product or modify the internal structure by yourself to avoid malfunction or accidents such as electric shock hazard. - Do not touch the device with wet hands or unplug the power cord. Otherwise, it may cause electric shock. - Do not touch the lens and the LED part directly with your fingers to avoid danger or damage to the lens. - If the lens framing is dirty and damp, it is best to use a dry, linen-free cloth or professional lens test paper to wipe it.

Chapter 2: Precautions for safe use

2.1 Safe use period

The safe use period is five years and the date of manufacture is shown on the product label.

2.2 Safe Use

In accordance with the electrical safety classification, this equipment is classified as Class I, Portable General Purpose Equipment.

The following is an overview of safety precautions:

1. Do not open the housing of the device under any circumstances by the operator.
2. Do not disable or short-circuit the safety function components of the built-in equipment.
3. This equipment is portable, the equipment is strictly prohibited to move when in working condition.
4. If the equipment can not work properly do not implement their own maintenance, should be immediately with the company qualified and authorised to maintain the equipment of the personnel contact.
5. The machine does not have the repair parts you need inside.
6. Obey all warnings and cautions, whether clearly stated or unspoken.
7. Follow all safety labelling on the equipment.

2.3 Prone to incorrect use or operation

Eyes open during photo taking leads to blinding phenomenon.

2.4 Emergency Measures for Unusual Situations

People who are allergic to fluorescent lights or UV light sources should stop taking pictures.

2.5 Considerations for Special Situations

1. Sudden power failure, disconnect the machine from the power supply after power failure.
2. Unplug the power supply before moving the equipment.

2.6 Contraindicated population

Use with caution in cases of light sensitivity.

2.7 Symbols and labels

2.7.1 safety symbol

	protective earthing
	AC

2.7.2 Transport symbols

	Fragile, handle with care
	Keep dry
	Stacking
	Position up

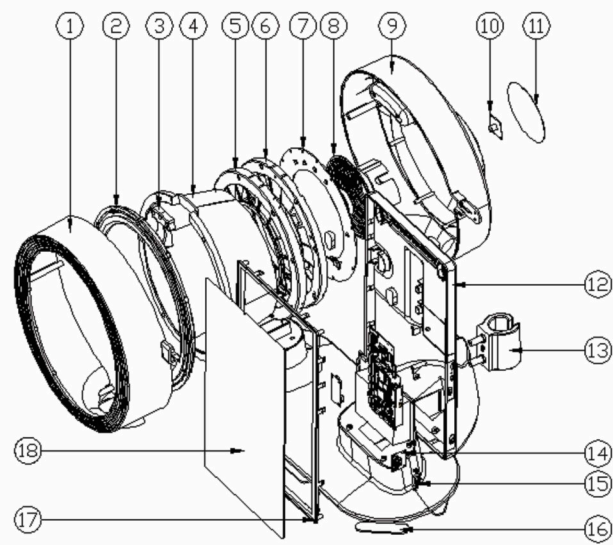
2.7.3 System labels

	Date of manufacture
	Serial number
	Refer to instructions for use

Chapter 3:
Structure and Working Principle Diagram

3.1 General structure and working principle

3.1.1General structure diagram:



No.	Name	Quantity	Remarks
1	Front shell	1	
2	Decorative ring	1	
3	Forehead pad	1	
4	Magic mirror-reflector	1	
5	Magic Mirror-Luminous Panel-Cover2	1	
6	Magic Mirror-Luminous Panel - Cove	1	
7	Light board	1	
8	Magic Mirror-Lens-Cover-1	1	
9	Backshell	1	
10	Camera	1	

11	Aluminium back cover	1	
12	Display Backshell	1	
13	Handle Hanger	1	
14	Pushbutton-PCB	1	
15	Pushbutton	1	
16	Silicon pad	6	
17	Display frame	1	
18	Display Screen	1	

3.1.2 Working Principle:

Intelligent facial photo instrument is one of the facial skin image data statistics instruments, can be in the RGB + UV + PL tri-spectrum to shoot the face of high-definition images and artificial intelligence AI algorithms to analyse the skin characteristics, so as to provide data reference for the beauty industry personnel to customize skincare solutions, and at the same time with the skin magnifying glass, you can be more detailed observation of the skin local conditions.

3.2 Workflow of main components or functional units

Workflow: Run the intelligent photo software in the tablet, the controller receives the tablet software instruction later to control the constant current source to control the corresponding LED lights, at the same time, the tablet is connected to the camera through the USB port and collects images for analysis.

Chapter 4: Technical Characteristics

Electrical power: 25 W	Self-display: 15.6 inch
Rated power: 110~230 VAC/50Hz~60Hz	Running memory: 2GB
Pixel: 20 million	Storage space: 64GB
Spectrums: 10 Spectrums	Photo Mode: Auto Mode
Running platform: Android 8.1.0	External Expansion: Support external SD card

Chapter 5: Size & Weight

Instrument size: Length 60.6CM*Height 45.5CM*Thickness 22.1CM
Packing size: L70CM*W55CM*TH 29.5CM
Gross weight: 9KG
Net weight: 5KG

Chapter 6: Installation and Commissioning

6.1 Installation Method

Open the packaging box, take out the device, and check whether the outer casing and accessories have any visible deformation. Verify that all accessories are complete according to the packing list.

6.2 Installation Precautions

- 1. Do not perform installation while the device is powered on.
- 2. The device should be connected to an AC power supply of 110–230 VAC ±10%.

6.3 Installation Test Items

Check the camera functionality, RGB light, PL light, and UV light illumination. Also, check the magnifier functionality.

6.4 Preparation Before Trial Operation & Trial Operation

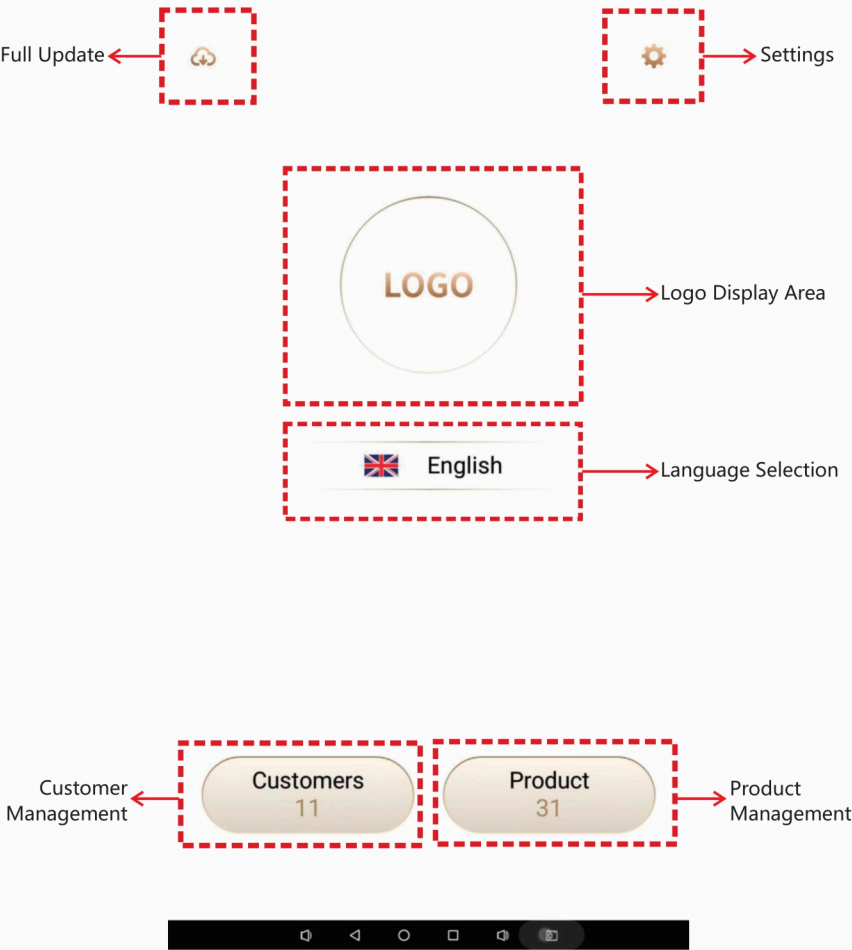
After connecting the device to the power supply, turn it on and follow the operation instructions to test whether all functions are working properly.

Chapter 7: Operating Guide

7.1 Interface Overview

7.1.1 Main Interface

Press and hold the button at the bottom right corner of the tablet to power on the device and enter the software interface (Picture 1-1). The main interface includes the following sections: Logo Display Area, Full Update, Settings, Customer Management, Product Management and Language Selection.



Picture 1-1

- LOGO:** Primarily used for in-store brand promotion.
- Customers:** Stores customer information and customer analysis records.
- Product:** Used to manage in-store products and recommend products based on analysis results.
- Settings:** The software settings allow for cloud registration, version updates, cloud product downloads, and backup & restore functions.
- Language Selection:** Click Language Selection to choose the applicable language.

7.1.2 Registering to an Institution:

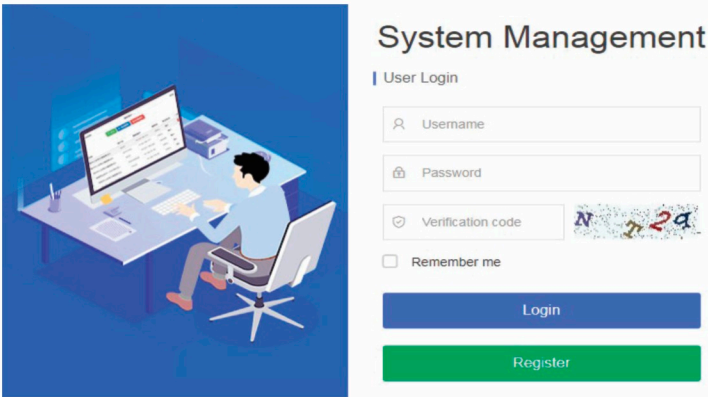
Cloud Account Registration and Device ID Input Method:

Registering to an institution is required when using the network functions of this device, such as uploading logo and product information. Each store is linked to its own institution for management.

Step 1: Ensure that the device is connected to the WiFi.

Step 2: Use a computer to register a cloud account.

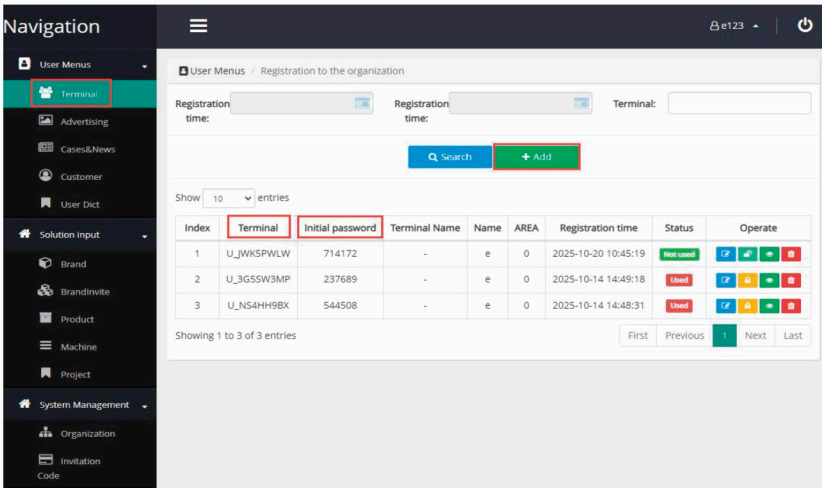
- Open a browser on the computer and go to: <http://analysis.milisun.com/login.aspx>
- The login page (as shown in Picture 1-2) will appear.
- Click **"Register"** at the bottom to create a cloud account.
- Fill in the registration form based on the store's actual details (Picture 1-3).



Picture 1-2

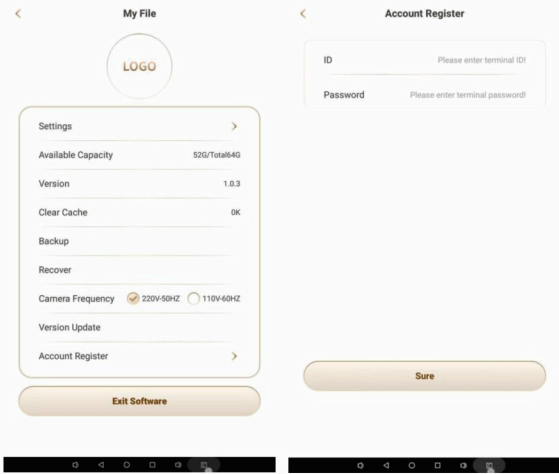
Picture 1-3

Step 3: Add an ID Account: Log in to the registered cloud account on a computer, then click **"Terminal"** to add a new ID account. (Picture 1-4)



Picture 1-4

Step 4: Entering the ID on the Device: On the device interface, there are four options: Customers, Product, Settings, and Full Update. Click **"Settings"**, then go to **"Account Register"**. Enter the previously registered **terminal ID** and **password** to bind the device to the cloud, enabling data synchronization and centralized management across multiple stores. (Picture 1-5).

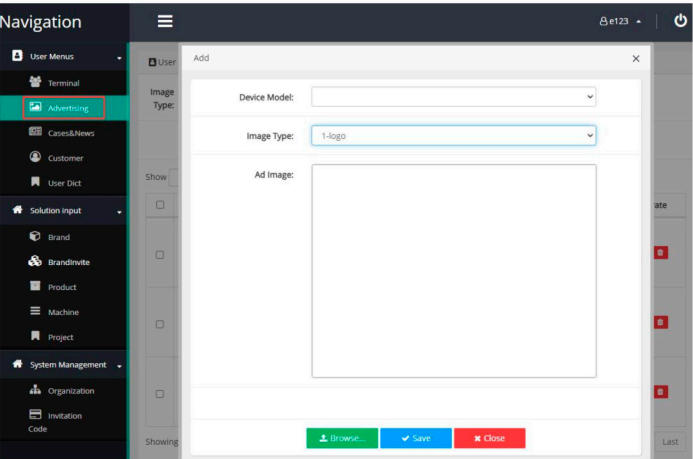


Picture 1-5

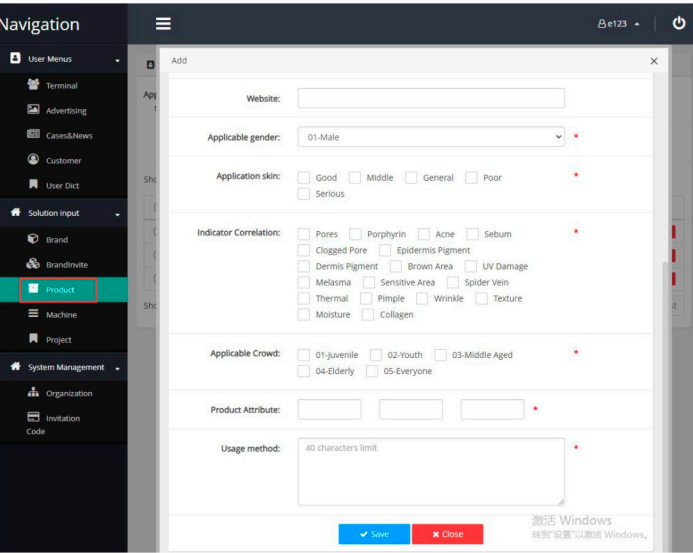
7.2 Data Upload and Download

7.2.1 Data Upload

This step allows you to upload information such as **logo, products and user dict**. You can use this function to add the required data. Method to Add a Logo (Picture 2-1). The method for adding products and user dict(conditioning suggestion, product attributes, standard value settings, product relationship, applicable skin) is the same. (Picture 2-2)



Picture 2-1



Picture 2-2

7.2.2 Data Download

Download options include **product, logo, standard value, conditioning suggestion, product function**. Click "Settings" icon, then go to "Settings" and select the corresponding item to download. (Picture 2-3)

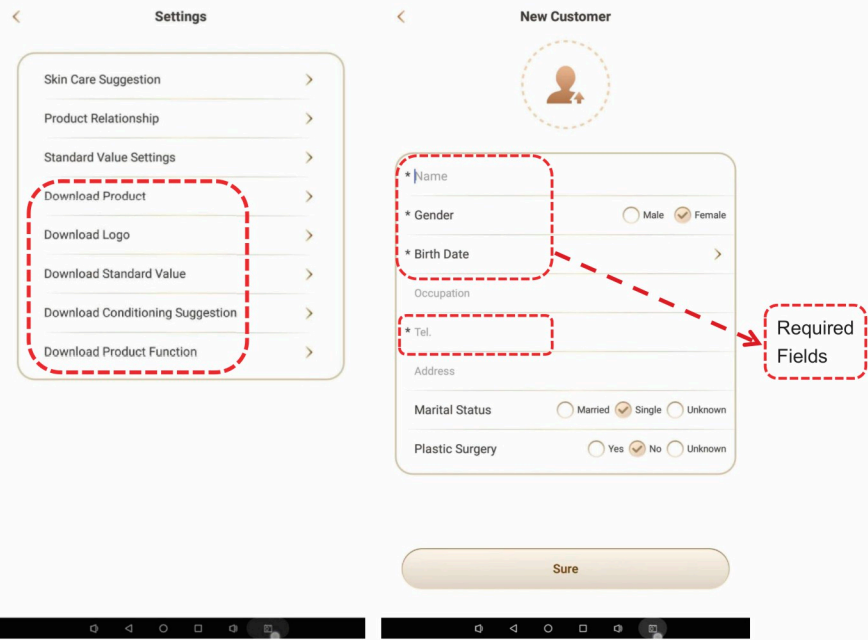
7.3 File Management

7.3.1 Create Customer File

*** Required Fields:** Name, Gender, Birth Date, Tel., Occupation, Address, Marital Status and Plastic Surgery can be filled out as needed. (Picture 3-1)

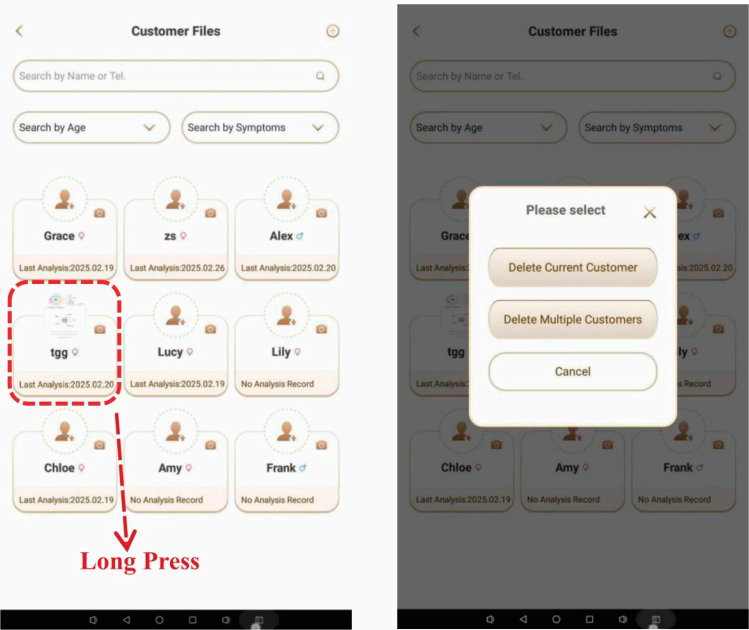
7.3.2 Delete and search files

Delete: In the main interface, enter the customer file section and locate the customer information you wish to delete. Long press the customer you want to remove, and the system will pop up a "Delete" button. Click Delete to remove the file. (Picture 3-2)



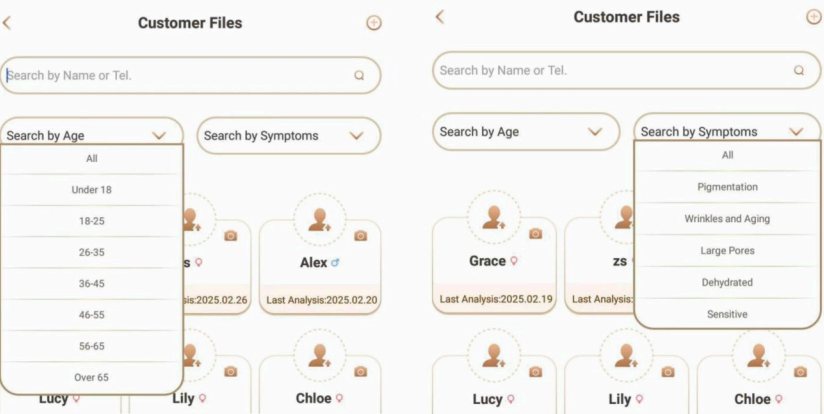
Picture 2-3

Picture 3-1



Picture 3-2

Search: In the main interface, enter the customer file section. The first method is to search by entering the customer's name or contact information. The second method is to search by age or symptoms. (Picture 3-3)



Picture 3-3

7.4 Photo capture and analysis

7.4.1 Photo capture and precautions

One-Click Shooting: Press the button once to take 3 consecutive shots, and the images will be immediately processed. The operation is simple and quick. (After taking the photos, the system will automatically detect the facial area based on the subject's face and avoid the facial features' positions.)

Photography Guidelines:

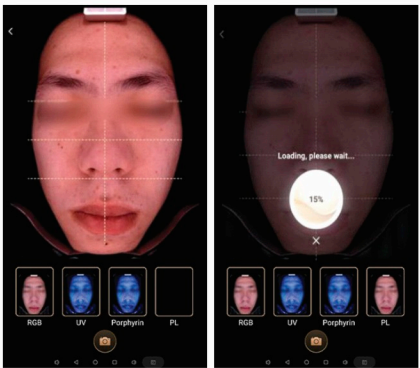
- **Hair Issues:** The face area must be clean and free from foreign objects and hair when taking the photo. Hair contains a large amount of pigment and may be misinterpreted as pigmented spots, affecting data accuracy.
- **Makeup Issues:** Photos should be taken with no makeup. Cosmetics may cover surface skin problems, leading to biased data and even obstructing the instrument's light, hiding deeper skin issues and affecting data accuracy.
- **Angle Issues:** The person being photographed should rest their chin on the chin rest and their forehead on the forehead rest, facing forward. The head should be adjusted properly to prevent issues like looking down or up, which could cause inconsistent facial regions and prevent data comparison, affecting data accuracy.
- **Light Blocking Issues:** When taking the photo, be sure to use a Hood to avoid external light sources affecting the shooting environment and causing inaccurate data analysis.
- **Hood Guidelines:** It must completely block the light to prevent interference from external light sources.

7.4.1.1 Photo confirmation and process

After taking the photo, choose either "Re-take Photo" or "Start Analysis" to confirm the current photo.

"Re-analysis" will allow you to take another picture.

"Precise Analysis" confirms that the current photo meets the requirements and proceeds to facial recognition for quantitative analysis. Press "Precise Analysis" to start the data analysis. (Picture 4-1)



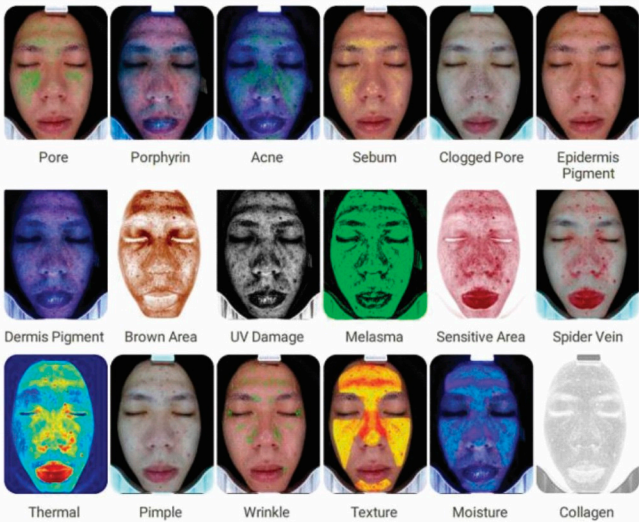
Picture 4-1

7.4.2 Indicator analysis and viewing

Through AI deep learning, the facial big data automatically selects the facial detection area, accurately avoids facial features, and continually improves its accuracy for quantitative skin analysis.

The top 10 deep skin issues are analyzed, and with precise data and real images, customers can clearly see each skin problem.

Each indicator's data is presented separately, allowing a clear view of localized facial issues, pore enlargement, and a more intuitive view. It also provides a deeper magnification to examine underlying issues and clearly view acne conditions. (Picture 4-2)



Picture 4-2

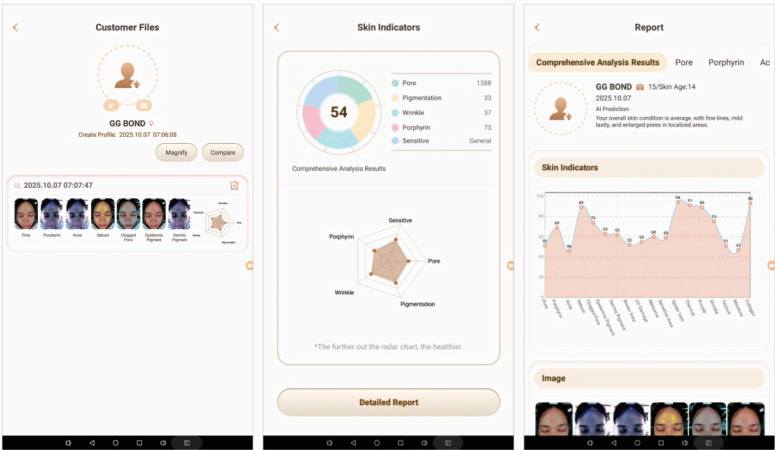
7.5 Report Viewing

7.5.1 Customer Report

Click on the image to enter the individual item report or the comprehensive report.

Comprehensive Analysis Report: The score-based quantitative analysis results are clear at a glance, along with skincare suggestions and a personalized beauty plan for the customer.

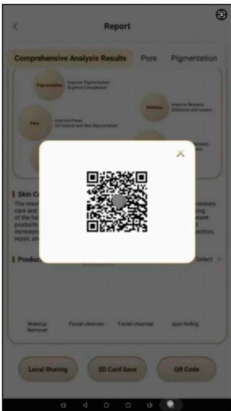
Individual Item Report: The data is quantified and provides an intuitive view of detailed problem data, skincare suggestions, and a personalized beauty plan for the customer. (Picture 5-1)



Picture 5-1

7.5.2 View report on mobile

Click "QR Code" and scan with your phone to view the test results on your mobile device.(Picture 5-2)

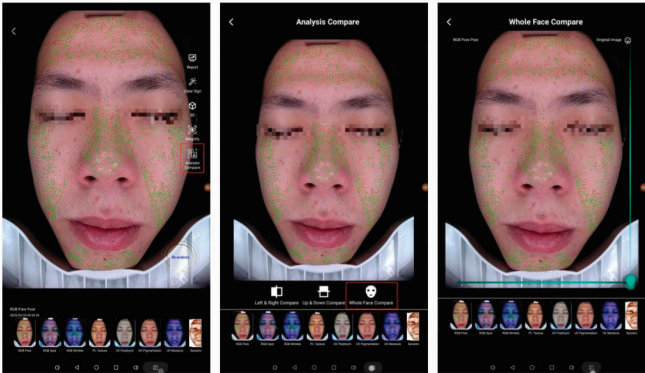


Picture 5-2

7.6 Analysis Compare

7.6.1 Whole Face Compare

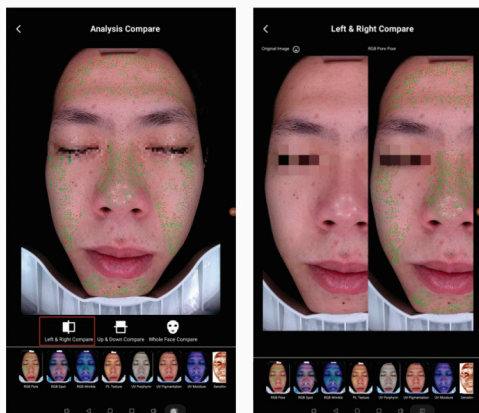
Click on the 'Analysis Compare' icon on the right side of the main interface, and slowly drag the green dot from the right side to the left side to perform a whole face comparison.(Picture 6-1)



Picture 6-1

7.6.2 Left & Right Compare

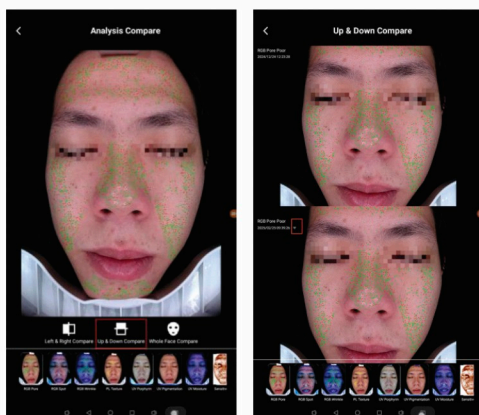
Click on the 'Left & Right Compare' icon, comparing two photos with high definition photo and 12 kinds of analysis from the same test by the left and right face. Slide from right to left could check all picture.(Picture 6-2)



Picture 6-2

7.6.3 Up & Down Compare

Click on the 'Up & Down Compare' icon, comparing two photos with high definition photo and 12 kinds of analysis from tests of different date by up and down face. Slide from down and up could check all picture. And it also can be used as a comparison of historical records from different periods of time. (Picture 6-3)



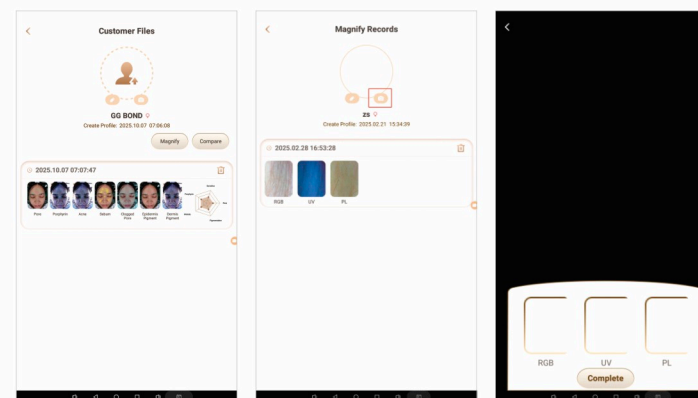
Picture 6-3

7.7 Magnify Analysis

Magnifier handle is an auxiliary device of skin analyzer, can be more clearly observed the skin local conditions, at the same time in the facial photo instrument inside, can be multi-position local shooting and record the shooting position, aspects of care before and after the control.

Operation method: Firstly, connect the equipped magnifier (the connection port is on the right side of the screen). After that, enter the local inspection interface of the customer file, click  and press the button on the handle of the magnifying glass to take photos.

(Picture 7-1)

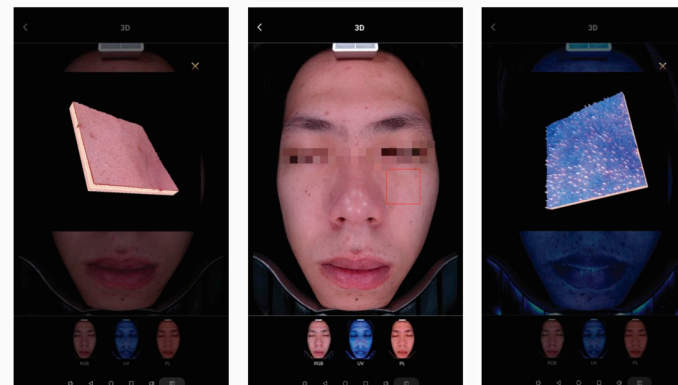


Picture 7-1

7.8 3D Slice

Click the '3D' icon, then drag the red tag to the area you want to view, and the 3D slice will be automatically displayed. This function can view slices of skin surface and deep problems.

(Picture 8-1)



Picture 8-1

7.9 Skin Aging

To predict the aging of your skin condition in the future, click the "Skin Aging" analysis item and select the "Prediction Year Timeline" icon to view the aging degree. (Picture 8-1)



Picture 9-1

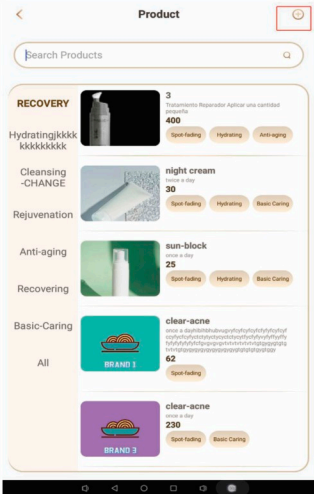
7.10 Product Management

7.10.1 Add Product

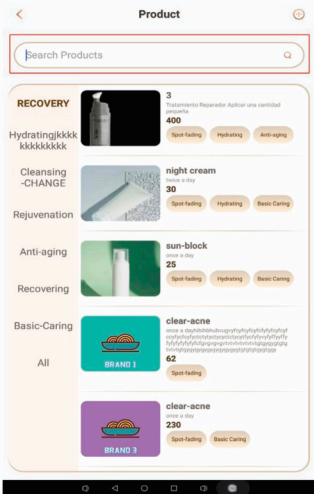
In the Product Suggestion, you can download products from the cloud or add new products locally.
Add products locally: Click the Add button and enter product information including image, brand, price, applicable gender, applicable skin, applicable crowd, product function, usage. (Picture 10-1)

7.10.2 Search Products

Click the search bar and enter keywords to find the products you want to add. (Picture 10-2)



Picture 10-1



Picture 10-2

7.10.3 Modify Product

On the product management page, select the product that needs to be modified, modify it, and save it. (Picture 10-3)

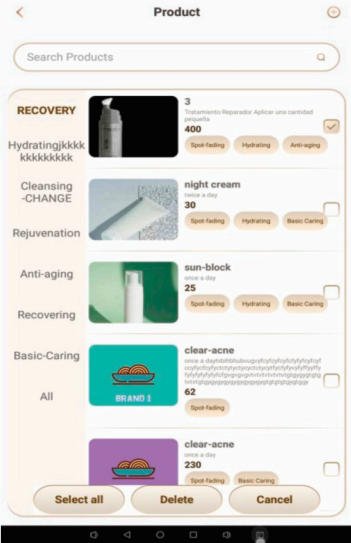
7.10.4 Delete Products

Long press any product to enter the management state, select (you can select in batches) the product and delete it. (Picture 10-4)

Product Management Form:

- * Product Image
- Brand Name
- * Product Name
- * Price
- Product Link
- * Applicable Gender: ☐ Male ☒ Female
- * Applicable Skin
- * Applicable Crowd
- * Product Function
- * Usage
- Sure

Picture 10-3



Picture 10-4

7.11 Update

In order to provide a better experience, the developer will periodically upgrade the software and fix related problems according to the problems encountered during use.
The specific operation is: After the current software is connected to WiFi, it will automatically detect whether there is an update after logging in with the password. You can also update manually. According to Picture 1-1, click the "Settings" icon, select "Version Update" to complete the update, and then log in to the software again.

7.11.1 Backup and Restore

Back up local data by inserting an SD card.

Chapter 8: Troubleshoot

No.	Symptom	Possible Reason	Method
1	Tablet does not boot	Power outlet loose	Please plug in the power line
		Power switch not turned on	Turn on the power switch
		The host is not turned on	Long press the host key to start device
2	Unable to upgrade software	Wi-Fi is unconnected	Exit software, connect Wi-Fi, upgrade again
3	Unable to login system	Forgot account name or password	Please contact the manufacturer to retrieve password
4	Shooting Crash	Interference by other aesthetic device	Turn off RF, Laser Interference device. Restart device again

Chapter 9:

Safety devices and accident handling

9.1 Installation of protective devices and precautions

This instrument needs to be installed on a socket with a leakage protector.

9.2 How to deal with accidents

If opening your eyes while taking a photo causes glare, just close your eyes in time.

9.3 Emergency handling methods

When the device malfunctions, disconnect the power supply and unplug the adapter.

Chapter 10: Maintenance And Repair

10.1 Use the instrument reasonably according to the operation steps indicated in the instruction manual, and protect the appearance of the instrument during use.

10.2 The instrument is not allowed to be turned on without normal operation.

10.3 During the use of the instrument, please remember to control the instrument with wet hands and adjust the operation mode of the instrument.

10.4 After the use of the instrument, please confirm the normal closure, and then wipe clean, regular spot checks on the instrument operator technician, whether there is a violation of the provisions of the operating procedures.

10.5 The product since the date of sale within one week of the quality of non-human factors, the command pany is responsible for the package return, and package replacement.

10.6 Warranty: Under normal use and storage, this product will be given free maintenance if quality problems occur within one year from the date of sale.

The following cases are not covered by the warranty:

- 1.Damage or deformation of the whole machine due to collision.
- 2.The whole machine into the water or rain.
- 3.Their disassembly, repair, and remodeling of products caused by the failure.
- 4.Not according to the instructions of the correct method of operation caused by the failure.
- 5.Unforeseen natural disasters (such as fire, earthquake, flooding, etc.) caused by the damage.

Chapter 11: Transportation And Storage

11.1 This instrument is suitable for general transportation by train, automobile, ship, airplane and so on.

11.2 The instrument should be prevented from violent vibration, collision and falling during transportation, and should be upward, lightly placed and protected from rain.

11.3 The instrument should be stored in a room without corrosive gas and well ventilated.

11.4 When the instrument is not used for a long time, it should be cleaned and stored, and taken out and energized once a year to prevent moisture.

Chapter 12: Unpacking And Inspection

- 12.1 Check whether there is any damage to the outer package before opening the box, if there is any damage, you need to contact with the relevant departments and take photos for documentation. Under the permission of the relevant departments, the box can be opened. If the instrument is damaged, we can go through the claim procedure with the transportation department.
- 12.2 Verify whether the packing list matches with the real thing in the box after opening the box (compare with the packing list), if there is any shortage, contact with the sales department of the instrument immediately.

Chapter 13: Environmental And Others

This product will not cause any impact on the environment after it is scrapped and no special treatment is required. Customers can dismantle and classify it for scrapping on their own.

Packing List

Product name	Quantity	Options (√)	Note
Main Body	1	√	
User Manual	1	√	
Certificate	1	√	
Warranty Card	1	√	
Power Cable	1	√	
Magnifier	1	√	
SD Card	1	√	